

Stephanie Wang

Education

Ph.D. and M.S. in Mathematics, *UCLA*, Eugene V. Cota-Robles Fellow. **2014-2020**
B.S. in Mathematics, *National Taiwan University*, *magna cum laude*. **2009-2013**

Experience

Research Engineer – reporting to Dr. Qingnan Zhou, *Adobe*, Seattle, WA. **2024-2025**
Geometry and physics research and software development, with a focus on geometry processing in machine learning and 3D modeling. Mentored students: [Yutao Zhang](#), [Baptiste Genest](#), [Dylan Rowe](#).
Postdoc – with Prof. Albert Chern, *UCSD*, San Diego, CA. **2020-2023**
Geometric deep learning, geometry optimization and physics-based simulation of various materials (solids, fluids, waves, cloth, hair, frictional contact, plasticity, fracture, *etc.*). Mentored students: [Mohammad Sina Nabizadeh](#), [Shiyang Jia](#), [Chad McKell](#), [Hang Yin](#), [Baichuan Wu](#).
Assistant Adjunct Professor / Instructor, *UCLA*, Los Angeles, CA. **2019-2020**
Taught three courses: linear algebra, machine learning (remote) and multivariable calculus (remote).
Ph.D. Study – with Prof. Wilfrid Gangbo, *UCLA*, Los Angeles, CA. **2019-2020**
Regularity theory for minimizers of polyconvex functionals related to Navier-Stokes equation.
Exchange Study – with Prof. Johan Gaume, *EPFL*, Lausanne, Switzerland. **2019 summer**
Simulating snow and tire interaction using physics-based simulation, in collaboration with Michelin tires.
Ph.D Study – with Prof. Joseph Teran, *UCLA*, Los Angeles, CA. **2016-2019**
Physics-based simulation of various materials (fracture, cloth, hair, deformable body, *etc.*) using Material Point Method and Finite Element Method.
Technical Intern, *Walt Disney Animation Studio*, Burbank, CA. **2018 summer**
R&D for pioneering simulation technology in animated feature films, teaming with VFX artists.
Teaching Assistant, *UCLA*, Los Angeles, CA. **2015-2020**
TA-ed courses: linear algebra, numerical methods (intro, intermediate, and advanced), C++ programming (intro, intermediate, and advanced).

Skills

Programming: C++, Python, MATLAB, $\LaTeX$, Vim, git, zsh, Houdini
Math: Optimization, differential geometry, solid and fluid dynamics, scientific computing
Languages: English and Mandarin Chinese (bilingual)

Selected Publications

[Implicit UVs: Real-time Semi-global Parameterization of Implicit Surfaces](#). Computer Graphics Forum (Eurographics 2025).
[Baptiste Genest](#), [Pierre Gueth](#), [Jérémy Levallois](#), [Stephanie Wang](#).
[Exterior Calculus in Graphics: Course Notes for a SIGGRAPH 2023 Course](#). SIGGRAPH 2023.
[Stephanie Wang](#), [Mohammad Sina Nabizadeh](#), [Albert Chern](#).
[DeepCurrents: Learning implicit representations of shapes with boundaries](#). CVPR 2022.
[David Palmer](#), [Dmitriy Smirnov](#), [Stephanie Wang](#), [Albert Chern](#), and [Justin Solomon](#).
[Computing minimal surfaces with differential forms](#). ACM ToG (SIGGRAPH 2021).
[Stephanie Wang](#) and [Albert Chern](#).
[Simulation and visualization of ductile fracture with the material point method](#). PACM-CGIT (SCA 2019 **Best Paper Award**).
[Stephanie Wang](#), [Mengyuan Ding](#), [Theodore F. Gast](#), [Leyi Zhu](#), [Steven Gagniere](#), [Chenfanfu Jiang](#), and [Joseph M. Teran](#).
...and 7 more papers published in top journals in computer graphics and other scientific fields.